1. Forecast design and source audit

Objective: forecast Botswana FAO Item 23014 food-price inflation for January-December 2024 using only information available through December 2023.

Five-source integration

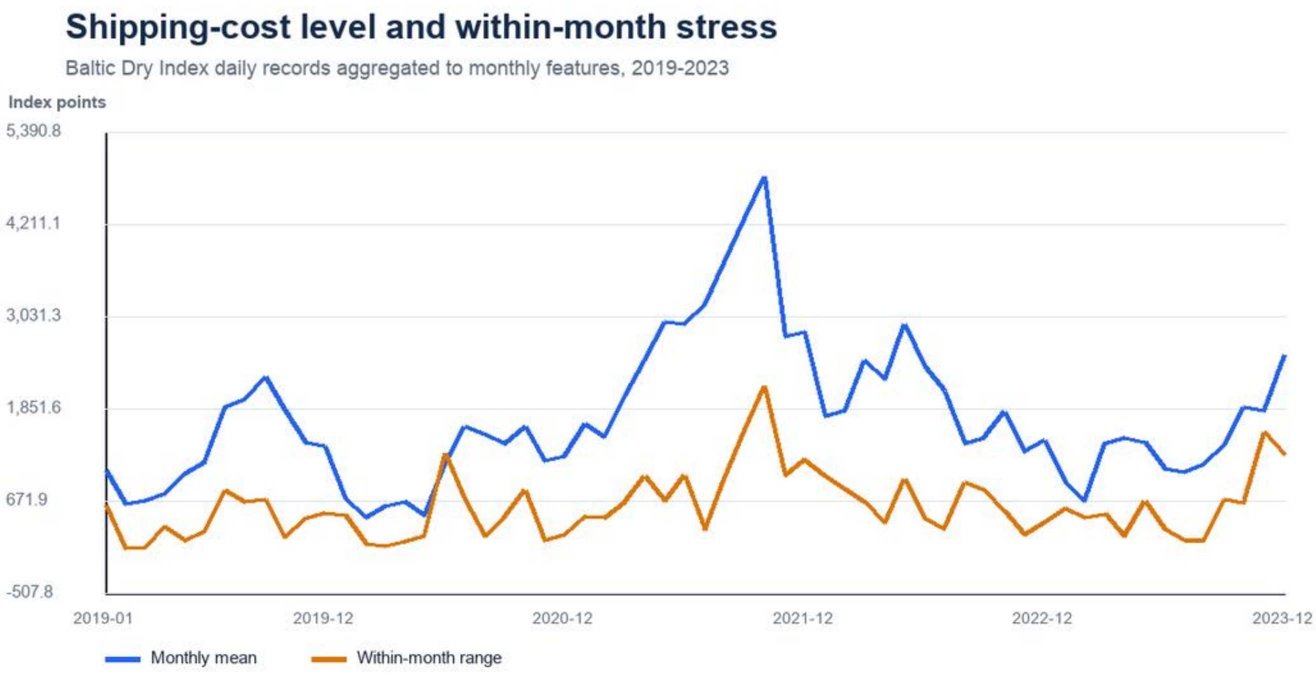
Source	Frequency	Rows	Role
Baltic Dry Index	Daily	5992	Shipping pressure
Brent crude	Monthly	288	Fuel and transport
Botswana policy rate	Monthly	288	Lagged policy response
Botswana FAO prices	Monthly	852	Target and domestic indices
Cross-country FAO	Monthly	4320	Regional pressure signals

- Coverage is 2000-01-01 to 2023-12-01; Item 23014 provides 276 non-null monthly targets.
- Item 23014 reconciles to the 12-month percentage change in Item 23013; maximum absolute source discrepancy is 0.00000278 percentage points.
- Three exact daily shipping duplicates were removed in memory; raw competition files remain unchanged.
- All dates were normalised to month starts before outer joining onto a complete monthly calendar.

2. Daily shipping data to monthly signals

A monthly mean alone hides supply-chain stress. The pipeline derives level, dispersion, timing, momentum and tail-event features from each month's daily observations.

- Level: mean, median, minimum, maximum and range.
- Volatility: standard deviation, coefficient of variation, mean and maximum absolute daily move.
- Momentum: first-to-last return and linear within-month trend per trading day.
- Timing: second-half minus first-half average, preserving late-month acceleration.
- Extremes: counts of daily gains at or above 3% and losses at or below -3%.
- Market microstructure: trading-day count and mean high-low spread ratio.



3. Lag structures and economic rationale

Observable-at-origin design

The pipeline constructs 141 origin-safe features. The GRU consumes a compact cross-dataset channel set, while SARIMA provides a univariate target-history baseline. Current values are allowed only when published by the forecast origin.

Variable family	Transformations	Economic rationale
Food inflation	lags 0-24; rolling 3/6/12/24; momentum	Persistence and annual base effects
Domestic indices	lags 0/1/3/6/12; changes; rolling means	Price-level transmission
Shipping	rich aggregates plus lags 0/1/3/6/12	Import-cost pass-through
Brent	levels, changes, rolling mean and lags	Fuel and transport costs
Policy rate	levels and lags through 12 months	Delayed monetary response
Regional prices	country lags, rolling means and index growth	SACU/trading-partner pressure

South Africa and Namibia are prioritised because of Botswana's trade and customs links; Kenya and Zimbabwe provide robustness across different regional inflation regimes.

4. Missing futures, leakage controls and limitations

No phantom 2024 features

The GRU directly emits all 12 horizons from the December 2023 sequence; SARIMA produces a 12-step endogenous forecast from target history. Neither model forecasts, imputes or treats 2024 Baltic Dry Index, Brent or policy-rate values as known.

- Training rows are accepted only when their complete 12-month label window ends before the fold cutoff.
- Scaling parameters are estimated separately inside every rolling training fold.
- SARIMA specification tuning uses 2018-2020; final model comparison uses untouched 2021-2023 calendar-year backtests.
- The final January-December 2024 forecast is generated from the December 2023 origin.

Known limitation

The supplied file named Human Capital Project contains FAO price indicators for five countries, not direct education, health or nutrition outcomes. These series are used as regional price-pressure features; the linkage memo does not claim direct human-capital causality.